

Accounting for the role of investment frictions in recessions

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Abstract

Our business cycle accounting exercise reveals that both capital and investment efficiency declines played a prominent role in accounting for the output downturn during the US Great Recession. The evidence indicates that an increase in firms' investment costs may have played a substantial role during the US Great Recession, consistent with business cycle models in which firms face financial frictions. The negligible role played by the total factor productivity decline in accounting for the output downturn during the US Great Recession found by previous works can be explained by the movement in opposite directions of both labour and capital efficiency. However, we find that labour efficiency falling was the main force driving output downturn in the 1982 recession and the euro area Great Recession.

1 | INTRODUCTION

The business cycle accounting (BCA) method developed by Chari *et al.* (2002,2007) uses the equilibrium conditions of the dynamic stochastic general equilibrium (DSGE) model to measure the wedges representing the overall distortions to the relevant equilibrium conditions of the model. Usually, four wedges are computed (labour, investment, efficiency and resource constraint) and then fed back into the model one at a time to assess how much of the observed movements of macroeconomic variables can be attributed to each wedge. The BCA method allows us to reveal the mechanisms through which fundamental processes drive economic fluctuations. Therefore, it is a methodology for determining the most promising kinds of theories regarding the primary characteristics of economic fluctuations, and has been used in many countries and periods (see, for example, Brinca 2013, 2014; Cavalcanti 2007; Chakraborty and Otsu 2013; Chakraborty 2006; Cho and Doblas-Madrid 2013; Kersting 2008; Kobayashi and Inaba 2006; Otsu 2010; Rodríguez-López and Solís-García 2016; Sustek 2011). Brinca *et al.* (2020) provide a summary of this literature.

The objective of this paper is to carry out a BCA exercise for the two main recent US recessions (the 1982 recession and the Great Recession) and the Great Recession in the euro area (formed

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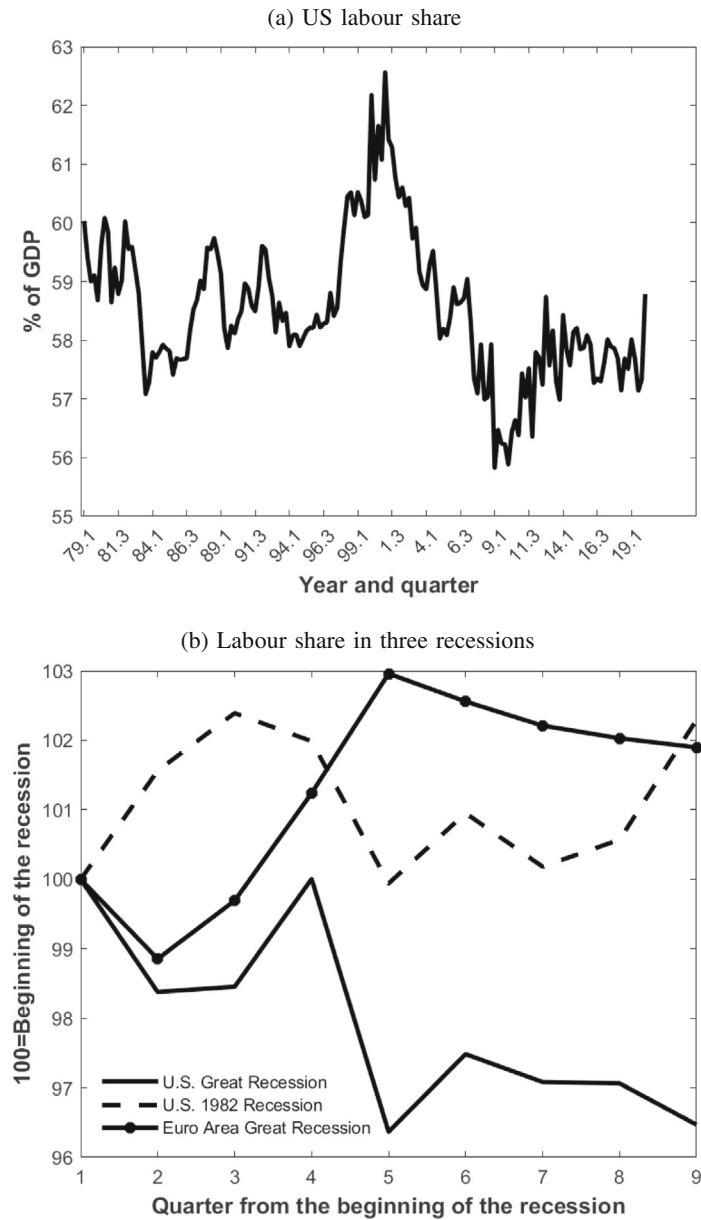
by 12 countries) with the aim of accounting for the role played by the efficiency wedges of labour and capital as well as discussing the relative importance of the investment and labour wedges. We perform our exercise under two alternative assumptions: considering that either (i) the marginal productivities of factors equal the factor rental prices and thus the labour wedge reflects only the gap between the real wage and the marginal rate of substitution between consumption and leisure, or (as argued by Karabarbounis 2014) (ii) the marginal productivities of factors differ from the factor rental prices due to market frictions or non-competitive forces (as is assumed implicitly by Chari *et al.* 2002, 2007; Brinca *et al.* 2016) and thus, in addition, the labour wedge also reflects the gap between the marginal productivity of labour and the real wage.

We must underscore that we modify the Chari *et al.* (2007) BCA method, including in the analysis the factor income distribution. Therefore, unlike previous BCA exercises, we can compute two efficiency wedges, one for labour and another for capital. To compute both wedges, we make the model consistent with the trend and cyclical behaviour of the factor income shares. Computing two efficiency wedges can be very relevant empirically. If a single efficiency wedge is computed, then its movements might conceal very different movements in the capital-efficiency and labour-efficiency wedges. Therefore the same behaviour of the efficiency wedge might be explained by different mechanisms and theories. In particular, movements in the efficiency wedges of capital and labour might go in opposite directions and cancel each other out. In this case, we might conclude that changes in the efficiency wedge are not relevant in accounting for economic fluctuations, even if the changes in both the labour-efficiency and capital-efficiency wedges were playing a very active role. Indeed, we find that it might have been the case during the US Great Recession.

We assume that the marginal productivities of factors equal their rental prices to compute both efficiency wedges. Under this assumption, movements in the factor income shares are led by the frictionless competitive adjustment of the productive factors. However, Chari *et al.* (2007) assume implicitly that the factor rental prices differ from the marginal productivities of factors. Moreover, they assume a Cobb–Douglas (CD) production function. Therefore movements in the factor income shares are led by market frictions or non-competitive forces moving the factor rental prices away from the marginal productivities of factors. This means that the labour and investment wedges computed by Chari *et al.* (2007) reflect movements in the factor income shares. This is not the case with the investment and labour wedges computed using our method (we clarify this point in the next section). That is not a mere theoretical curiosity because, from the 1950s, the US labour share underwent a significant decline as well as important oscillations (see Figure 1(a)). Moreover, the labour share underwent a significant decrease during the US Great Recession, but it increased during the euro area Great Recession and the US 1982 recession (see Figure 1(b)). Indeed, we find that assuming that either the movements in the factor income shares are due to market frictions and non-competitive forces, or the frictionless competitive adjustment of the productive factors has significant consequences for the computation of the labour and investment wedges and their effects for the economic fluctuations.

Although many authors find evidence of a decline in the labour share in the USA and other countries, there is some discussion on the subject. Elsby *et al.* (2013) show that the labour share decreased in most US industries during 1987–2011. In the same vein, del Río and Lores (2019) show that the US labour share fell by 4.99% during 1998–2015. Moreover, the labour share in the US private sector (i.e. excluding the government sector) fell by 6.25%. During this period, del Río and Lores (2019) find that the labour share fell in 40 of 60 industries of the US private sector. Oberfield and Raval (2021) find that the decline in labour share for the US manufacturing sector was higher than the fall of the labour share in the whole economy. However, Gomme and Rupert (2004) find that the labour share of the US non-financial corporate business sector and the labour share of the US economy excluding government and housing sectors remained roughly stable from the end of the 1970s to the early 2000s. Rognlie (2015) argues that the increase in the US net capital share from the mid-1980s turns out to come entirely from the housing sector, and

FIGURE 1 The labour share.



Koh *et al.* (2020) argue that intellectual property capital accounts entirely for the decline in US labour share, which is secularly constant for structures and equipment capital. Lawrence (2015) argues that the decline in the ratio of effective capital to effective labour in a context of biased technical progress and gross complementarity between capital and labour can account for much of the recent fall in US labour share at both the aggregate and industry levels. However, Karabarbounis and Neiman (2014) argue that the decline in the relative price of investment, which induces capital deepening in a context in which capital and labour are gross substitutes, explains roughly half of the observed decline in labour share. Some authors find that the decline of the labour share is not confined to the USA. For OECD countries, Jones (2003) reports empirical evidence of an increase in the capital shares, and Cho *et al.* (2017) find that the labour share decreased significantly after the mid-1970s. Karabarbounis and Neiman (2014), using a dataset of 59 countries,

find that the labour share in the corporate sector declined significantly in the majority of countries since the early 1980s. Rodriguez and Jayadev (2010) estimate a declining average trend in labour shares using an equally weighted set of 129 countries.

We specify a DSGE model with both the CD and variable elasticity of substitution (VES) production functions to perform our BCA exercises. A VES production function such as the one used in this work arises from a constant elasticity of substitution (CES) production function with variable utilization of capital (see Section III). Both VES and CES production functions are estimated by del Río and Lores (2019) for the US economy. In both cases, they estimate that capital and labour are gross complements. Here, we use their estimates. Although there is some debate, empirical evidence indicates increasingly that the elasticity of substitution between capital and labour is lower than 1. Based on a literature survey, Chirinko (2008) concludes that the weight of the evidence suggests an elasticity of substitution in the range 0.4–0.6. Our parametrization of the VES production function implies that the elasticity of substitution between capital and labour is around 0.7.

In the VES case, we assume that the marginal productivities of factors equal the factor rental prices. Therefore the factor income shares are led by the frictionless competitive adjustment of factors. This assumption has two implications. First, it allows us to compute two efficiency wedges: one for capital, and another for labour. Second, the labour wedge reflects only the gap between the marginal rate of substitution between consumption and leisure and the real wage. To this regard, we follow del Río and Lores (2021), who use a VES production function to compute two efficiency wedges in addition to the resource constraint, labour and investment wedges in a neoclassical growth model to analyse US economic growth after the Second World War.

In the standard CD case, a single efficiency wedge can be computed. Moreover, since output elasticities for factors are constant, changes in the factor shares must necessarily be driven by market frictions or non-competitive forces moving the factor rental prices away from their marginal productivities. Therefore in the CD case, the labour wedge reflects the gap between the marginal productivity of labour and the real wage in addition to the gap between the marginal rate of substitution between consumption and leisure and the real wage. However, Karabarbounis (2014) argues that explanations of the labour wedge based on departures of the representative firm's marginal productivity of labour from the real wage are rejected by data because the labour share of income is not strongly procyclical. In particular, he concludes:

As a result, models that generate volatile and countercyclical labour wedges by modifying the firm side of the neoclassical growth model are rejected by the data. The most promising explanations of the labour wedge should be able to generate large deviations between the real wage and the household's measured marginal rate of substitution. (Karabarbounis 2014)

We compare the results obtained with both specifications. The CD specification is the standard case in the literature. In particular, it is the specification used by Chari *et al.* (2007). Therefore the wedges computed in the CD case correspond to the wedges computed with the Chari *et al.* (2007) method.

If the observed changes in the factor income shares reflect the frictionless competitive adjustment of productive factors, then we find that during the US Great Recession, (i) the capital-efficiency wedge was the main force driving the output growth slowdown, (ii) the labour and investment wedges were significant, but secondary forces, and (iii) the countercyclical evolution of the labour-efficiency wedge helped to reduce output growth slowdown.¹ However, we find that in the US 1982 recession and the euro area Great Recession, (i) the labour-efficiency wedge was the main driving force of the output growth slowdown, (ii) the labour was a significant, but secondary force, and (iii) the investment wedge was a negligible force.

Brinca *et al.* (2016) find that the efficiency wedge played a negligible role in the US Great Recession. Arellano *et al.* (2019) build a model accounting for the output growth slowdown during the US Great Recession, consistent with the Brinca *et al.* (2016) findings. They develop a model in which a deterioration of the financial frictions is related to a worsening of the labour wedge and downturns in aggregate labour and output, with small movements in total factor productivity (TFP). However, our results reveal that the Brinca *et al.* (2016) result might be due to the evolution of the efficiency wedge concealing movements of both the capital-efficiency and labour-efficiency wedges going in opposite directions and cancelling each other out. Moreover, as argued in Section II, financial frictions raising the firm's cost of investment can be manifested in either the capital-efficiency wedge or the investment wedge, or both. Therefore, in contrast to the results obtained by Brinca *et al.* (2016), our results suggest that the frictions in capital markets of the kind proposed by the *investment friction theory* might have been a prominent force driving the US Great Recession.² Chari *et al.* (2002) call the models with financial frictions raising the firm's investment costs and causing investment-driven downturns in output the *investment friction theory*. Some eminent examples of these models are by Carlstrom and Fuerst (1997), Kiyotaki and Moore (1997), Bernanke *et al.* (1999) and Gertler and Kiyotaki (2009).³

The remainder of this paper is organized as follows. Section II discusses the implications of the specification of the production function in measuring the wedges, the relationship between the wedges and the prices of capital assets, and the relationship between the investment costs and both the capital-efficiency and investment wedges. Section III describes the model. In Section IV, we describe the BCA procedure and the building of data. In Section V, we assess the role of the wedges in the three considered recessions. Finally, Section VI concludes.

2 | TECHNOLOGY, PRICES AND WEDGES

The BCA method developed by Chari *et al.* (2007) allows us to compute a single efficiency wedge reflecting TFP as well as estimating an autoregressive stochastic process of order 1 for it. There are infinite combinations of the labour-efficiency and capital-efficiency wedges consistent with the efficiency wedge. The Chari *et al.* (2007) method ignores the factor income distribution. However, introducing into the analysis an assumption on the factor income distribution (in particular, on the relationship between the output elasticities for factors and the factor shares) allows us to identify the particular combination of the labour-efficiency and capital-efficiency wedges consistent with the total efficiency wedge and the factor income distribution.

Consider the following neoclassical production function with constant returns to scale:

$$Y_t = A_t F(q_t K_t, z_t(1 + \gamma)^t H_t) = A_t z_t F\left(\frac{q_t}{z_t} K_t, (1 + \gamma)^t H_t\right). \quad (1)$$

The impossibility theorem of Diamond and McFadden (1965) states the impossibility of identifying simultaneously A , q and z . Chari *et al.* (2007) evade the impossibility by assuming that the efficiency of capital relative to labour is constant for all t (i.e. $q_t = z_t = 1$ for all t). They compute A_t from equation (1) using data for Y_t , K_t and H_t . The single efficiency wedge computed by Chari *et al.* (2007) reflects TFP. Here, we drop the technological assumption on the constancy of the efficiency of capital relative to labour. We assume that q_t/z_t is such that the output elasticity for capital equals the capital share:

$$\varepsilon\left(\frac{q_t}{z_t} \frac{K_t}{H_t}\right) = \varepsilon_{k,t}. \quad (2)$$

Then, normalizing $A_t = 1$, we compute q_t and z_t from equations (1) and (2) using in addition data for factor income distribution. Computing q_t and z_t requires departing from the CD specification

of the production function. The reason is that output elasticities for factors are constant. They do not depend on the ratio of effective capital to effective labour, $q_t K_t / z_t H_t$.

The single efficiency wedge computed with the Chari *et al.* (2007) BCA method is proportional to the weighted geometric average of both capital-efficiency and labour-efficiency wedges: $A_t = A_0 q_t^{\varepsilon_t} z_t^{1-\varepsilon_t}$, where A_0 is a constant. Consider the production function in equation (1) with $A_t = 1$, and alternatively with $q_t = z_t = 1$. Differentiating, it follows that the growth rate of TFP ($g_{A,t}$) is

$$g_{Y,t} - \varepsilon_t g_{K,t} - (1 - \varepsilon_t) g_{H,t} = g_{A,t} = \varepsilon_t g_{q,t} + (1 - \varepsilon_t) g_{z,t}.$$

Therefore, integrating, it follows that A_t is proportional to the weighted geometric average of the capital-efficiency and labour-efficiency wedges.

The single efficiency wedge might conceal very different behaviours of the capital-efficiency and labour-efficiency wedges. For example, if g_q and g_z are moving in opposite directions, then it might compute a low value for g_A , even if g_q and g_z are experiencing large variations. Therefore to calculate $g_{A,t}$ ignoring $g_{q,t}$ and $g_{z,t}$ might lead to the wrong conclusion that changes in the efficiency wedges of the factors are not significant in accounting for economic fluctuations.

Computing $g_{A,t}$ requires the output elasticities for factors. Chari *et al.* (2007) assume a CD production function. Then the output elasticities for labour and capital are constants ($\varepsilon_t = \varepsilon$), and

$$\tilde{g}_{A,t} = g_{Y,t} - \varepsilon g_{K,t} - (1 - \varepsilon) g_{H,t}.$$

Another alternative is to assume that marginal productivities of factors equal the factor rental prices. Therefore the output elasticities for factors equal the factor shares (i.e. $\varepsilon_t = \varepsilon_{k,t}$ and $1 - \varepsilon_t = \varepsilon_{h,t}$). In this case,

$$\bar{g}_{A,t} = g_{Y,t} - (1 - \varepsilon_{h,t}) g_{K,t} - \varepsilon_{h,t} g_{H,t}.$$

If the labour share ε_h goes down and the ratio of capital to worked hours is increasing, $g_{K,t} - g_{H,t} > 0$, then $\bar{g}_{A,t}$ decreases relative to $\tilde{g}_{A,t}$. Therefore assuming a constant output elasticity for labour might lead to underestimating the drop in the efficiency wedge, if the decline in the labour share is reflecting that output elasticity for labour is falling.

Moreover, as pointed out by del Río and Lores (2021), assuming constant output elasticities for factors might lead to overstating the fall of the labour wedge and to understating the fall of the investment wedge whether the labour share decreases. We focus on the labour wedge to clarify this point.

The first-order condition determining the time allocation of a household with additive log utilities of consumption c_t and leisure $1 - h_t$ is

$$\mu \frac{c_t}{y_t} \frac{h_t}{1 - h_t} = \pi_{h,t}^s \varepsilon_{h,t}, \quad (3)$$

where $\mu > 0$ is the value of leisure relative to consumption, y_t is output, $\pi_{h,t}^s$ is the labour supply wedge reflecting distortions in the labour supply, and $\varepsilon_{h,t} = w_t h_t / y_t$ is the labour share. The first-order condition for labour of a firm is

$$\pi_{h,t}^d (1 - \varepsilon_t) = \varepsilon_{h,t}, \quad (4)$$

where $\pi_{h,t}^d$ is the labour demand wedge, which reflects distortions of the labour demand, and $1 - \varepsilon_t$ is the elasticity of output for labour. Combining equations (3) and (4), we have the total

labour wedge

$$\pi_{h,t} \equiv \pi_{h,t}^s \pi_{h,t}^d = \frac{\mu}{1 - \varepsilon_t} \frac{c_t}{y_t} \frac{h_t}{1 - h_t}, \quad (5)$$

which reflects distortions in both the labour supply and demand. It follows from equations (3), (4) and (5) that the labour supply wedge $\pi_{h,t}^s$ does not depend on the shape of the production function, but the demand labour wedge $\pi_{h,t}^d$, and thus the total labour wedge $\pi_{h,t}$, does.

To compute the labour wedge, most authors (excepting del Río and Lores (2021), as far as we know) specify a CD production function and thus assume that the elasticity of output for labour, $1 - \varepsilon_t$, is constant. In this case, movements in the labour share are driven by changes in the labour demand wedge π_h^d (see equation (4)) and reflect frictions or non-competitive forces moving the rental prices away from their marginal productivities.

However, an alternative assumption is that output elasticity for labour equals the labour share (i.e. $1 - \varepsilon_t = \varepsilon_{h,t}$). In this case, rental prices and marginal productivities move together, thus changes in the labour share are driven entirely by competitive forces modifying output elasticities for factors (see equation (4)). Consequently, the labour wedge

$$\pi_{h,t} = \pi_{h,t}^s = \frac{\mu}{\varepsilon_{h,t}} \frac{c_t}{y_t} \frac{h_t}{1 - h_t} \quad (6)$$

exclusively reflects distortions of the labour supply (i.e. $\pi_{h,t}^d = 1$). Nevertheless, to implement this identifying assumption, the CD production function must be laid aside and we assume any other neoclassical production function with variable output elasticities for factors. Although the efficiency wedges do depend on the production function chosen, the labour and investment wedges computed in this second way do not.

In this paper, we compute the labour and investment wedges under both assumptions. It follows from equations (5) and (6) that the relationship between both labour wedges is

$$\pi_{h,t} = \pi_{h,t}^s \varepsilon_{h,t} / (1 - \varepsilon_t),$$

where $\pi_{h,t}$ is the labour wedge computed under the CD assumption and thus by Chari *et al.* (2002, 2007) and Brinca *et al.* (2016), whereas $\pi_{h,t}^s$ is the labour wedge computed under the VES assumption assuming that the marginal productivities of factors equal the factor rental prices (using a VES production function or any other with variable output elasticities for factors).

Therefore, when the labour share undergoes significant changes, the evolution of both labour wedges will differ significantly. In particular, if the labour share decreases (resp. increases), then $\pi_{h,t}$ decreases (resp. increases) regarding $\pi_{h,t}^s$.

Although it is not so evident, changes in the labour share (and then in the capital share) also affect the calculation of the investment wedge (see del Río and Lores 2021), just in the opposite sense to how they affect the labour wedge. In particular, if the labour share decreases (resp. increases), then the investment wedge computed assuming constant output elasticities for factors increases (resp. decreases) regarding the investment wedge computed assuming that the output elasticities for factors equal the factor income shares. To show this point, we assume that the depreciation rate of capital is 1, and there exist no adjustment costs of investment. The Euler equation of a perfect foresight model with additive log utilities of consumption and leisure implies that the investment wedge is given by

$$\pi_{x,t}^{-1} = \varepsilon_{t+1} \frac{y_{t+1}}{k_{t+1}} \frac{c_t}{c_{t+1}} \frac{\beta}{1 + \gamma}.$$

Brinca *et al.* (2016) assume a CD production function and then $\varepsilon_{t+1} = \varepsilon$ for all t . However, we assume that $\varepsilon_{t+1} = \varepsilon_{k,t+1}$. Therefore the relationship between the investment wedge computed under the former assumption (and by Brinca *et al.* 2016), $\tilde{\pi}_{x,t}^{-1}$, and the investment wedge computed under the latter assumption, $\bar{\pi}_{x,t}^{-1}$, is

$$\frac{\tilde{\pi}_{x,t}^{-1}}{\bar{\pi}_{x,t}^{-1}} = \frac{\varepsilon}{\varepsilon_{k,t}}.$$

If the depreciation rate of capital is lower than 1 or there exist adjustment costs of investment, then the previous relationship is only approximated.

We display the relationship between the wedges computed for the US economy under the CD assumption and the VES assumption in Figure 2. Consistently with the discussion above, the ratio is above (resp. below) the bisector during the Great Recession and below (resp. above) the bisector during the 1982 recession in the case of the investment (resp. labour) wedge (see Figures 2(b) and 2(c)). During the Great Recession (resp. 1982 recession), TFP computed in the CD case is higher (resp. lower) than the TFP computed in the VES (see Figure 2(a)). More interestingly, the relationship is horizontal during the US Great Recession, which means that in the CD case, TFP changed very little during the US Great Recession, much less than in the VES case.

Now we discuss the relationship between the capital-efficiency wedge, the investment wedge and the price of the capital assets. Consider that the capital services at time t are

$$Y_t = q_t K_t,$$

where q_t is the capital-efficiency wedge, and K_t is the stock of capital that evolves according to

$$K_{t+1} = b_t I_t + (1 - \delta) K_t,$$

where b_t is the efficiency of investment, and $0 < \delta < 1$ is the physical depreciation rate of capital. Under perfect competition, the price of capital assets at time t is

$$p_t = \frac{q_{t+1} b_t}{r_{t+1} + \delta_{e,t+1}},$$

where

$$\delta_{e,t} = 1 - (1 - \delta) \frac{p_{t+1}}{p_t} \frac{b_t}{b_{t+1}}$$

is the economic depreciation rate of capital. If $Y_t = C_t + I_t$ is the resource constraint, then both the evolution law of the stock of capital and the resource constraint can be rewritten in terms of the efficiency-adjusted investment, $X_t = b_t I_t$, as follows:

$$K_{t+1} = X_t + (1 - \delta) K_t$$

and

$$Y_t = C_t + \frac{1}{b_t} X_t,$$

which means that the investment wedge (which in this work is denoted by π_x^{-1}) at time t is $\pi_{x,t}^{-1} = b_t$. Therefore, according to the theory, the relative price of capital assets, p , is an increasing function of the capital-efficiency wedge and the investment wedge.

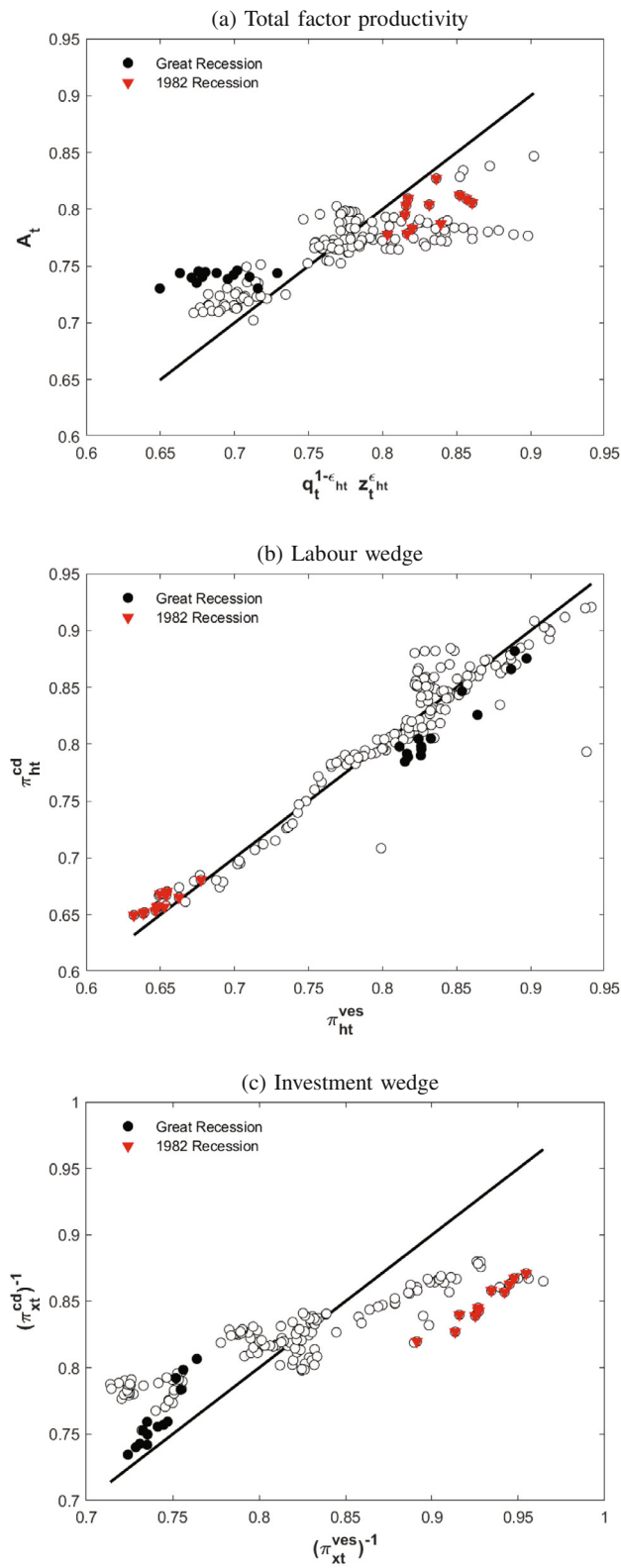


FIGURE 2 CD wedges versus VES wedges.

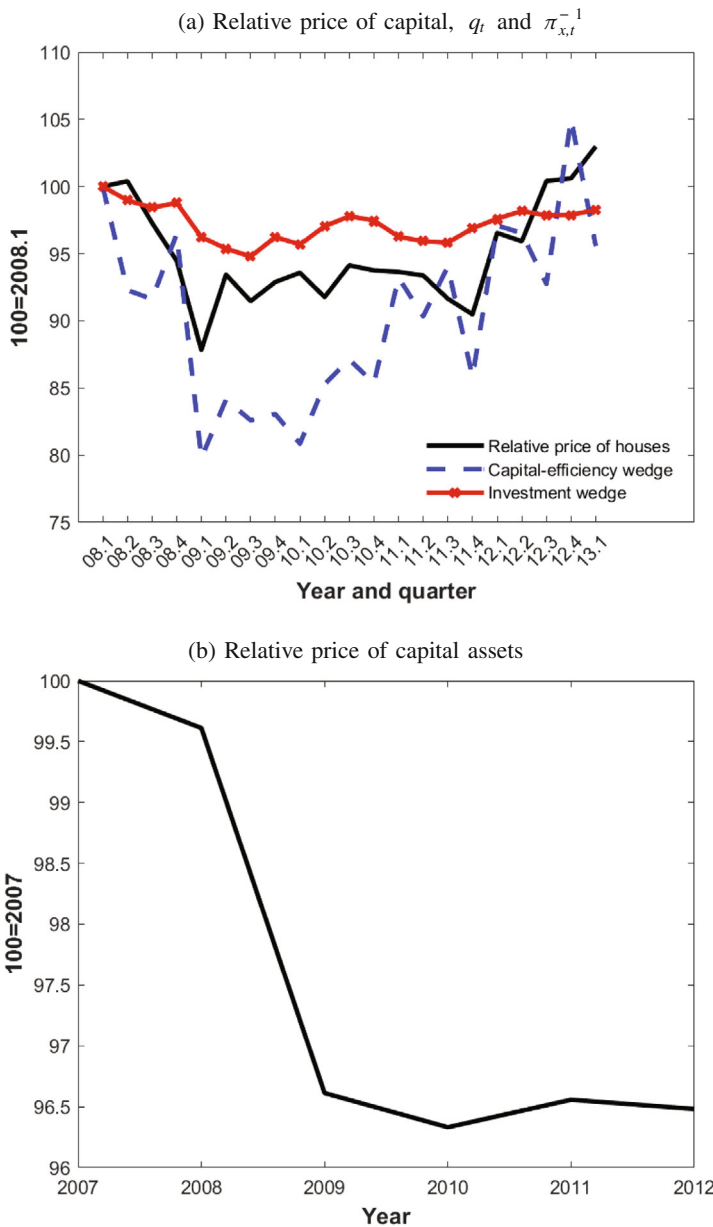


FIGURE 3 The relative price of capital, capital-efficiency wedge and investment wedge.

In Figure 3(a), we display the evolution of the US relative housing price during the US Great Recession, as well as the evolutions of the investment and capital-efficiency wedges computed below under the VES assumption. The evolutions of both wedges and the relative housing price are very related and consistent with theory. In Figure 3(b), we display the evolution of the relative price of US capital assets during the years of the Great Recession. Its drop is also consistent with theory and our computation of the wedges.⁴

As pointed out by Chari *et al.* (2002), most of financial frictions discussed by the *investment friction theory* end up affecting the economy by raising the firm's cost of investment, from 1 to $1 + \tau_x$. Chari *et al.* (2002) claim that these costs of investment show up as an investment wedge in the BCA exercise. However, this statement must be qualified. We argue below that if the costs linked to the financial frictions are not included in the measured investment, then the costs of

financial frictions show up as an investment wedge. Conversely, if the costs of financial frictions are included in the measured investment, then these costs show up as a capital-efficiency wedge.

Consider the resource constraint

$$C_t + (1 + \tau_{x,t})X_{1,t} = Y_t,$$

where C_t is consumption, $X_{1,t}$ is investment, $1 + \tau_{x,t}$ are the investment costs, and

$$Y_t = F(K_{1,t}, H_t)$$

is output, which is a function of labour H_t and capital $K_{1,t}$ that evolves according to

$$K_{1,t+1} = X_{1,t} + (1 - \delta)K_{1,t}.$$

If the measured investment is $X_{1,t}$ and the path of the stock of capital is built using $X_{1,t}$, then the investment wedge is $\pi_{x,t}^{-1} = (1 + \tau_{x,t})^{-1}$. However, if the measured investment is $X_t = (1 + \tau_x)X_{1,t}$ and the path of the stock of capital is built using X_t , then the three previous equations can be rewritten as

$$C_t + X_t = Y_t,$$

$$Y_t = F(q_t K_t, H_t)$$

and

$$K_{t+1} = X_t + (1 - \delta_t)K_t,$$

where $q_t = (1 + \tau_{x,t-1})^{-1}$ is the capital-efficiency wedge and reflects the costs of investment, $K_t = (1 + \tau_{x,t-1})K_{1,t}$ is the stock of capital adjusted by the costs of investment, and $1 - \delta_t = (1 - \delta)(1 + \tau_{x,t})(1 + \tau_{x,t-1})^{-1}$ is the economic depreciation rate of capital.

3 | THE MODEL

Our model is a neoclassical dynamic growth model with stochastic variables, henceforth called ‘wedges’. These wedges are ‘distortions’ and represent policies and institutions that affect productivity, hours worked, capital accumulation and resource constraint.

Output Y_t is allocated to consumption C_t , investment I_t , and other purposes G_t . A perfectly competitive representative firm produces output according to a neoclassical production function with constant returns to scale; it uses capital K_t and labour H_t as productive factors. Labour equals time worked per worker, h_t , times the number of workers (which equals population) L_t : $H_t = h_t L_t$. The number of workers (or population) grows at the constant rate η , where $L_{t+1}/L_t = 1 + \eta > 0$. Detrended output per worker, $y_t = Y_t / ((1 + \gamma)^t L_t)$, is given by

$$y_t = A_t f(q_t k_t, z_t h_t), \quad (7)$$

where $\gamma \geq 0$ is the rate of labour-augmenting technological progress, q_t is the *capital-efficiency wedge*, z_t is the *labour-efficiency wedge*, A_t is the (total) *efficiency wedge*, and $k_t = K_t / ((1 + \gamma)^t L_t)$ is detrended capital per capita.

The representative firm hires capital and labour to equalize its marginal productivities to their rental prices (r_t and W_t):

$$\varepsilon_t = r_t \frac{k_t}{y_t} \equiv \varepsilon_{k,t} \quad (8)$$

and

$$1 - \varepsilon_t = w_t \frac{h_t}{y_t} \equiv \varepsilon_{h,t}, \quad (9)$$

where $w_t = W_t/(1 + \gamma)^t$ is detrended wage per worked hour, ε_t is output elasticity for capital,

$$\varepsilon_t = \frac{f_1(q_t k_t/(z_t h_t), 1)}{f(1, z_t h_t/(q_t k_t))} \equiv \varepsilon \left(\frac{q_t k_t}{z_t h_t} \right), \quad (10)$$

and $1 - \varepsilon_t$ is output elasticity for labour. According to the first-order conditions (8) and (9), the capital share $\varepsilon_{k,t}$ equals the output elasticity for capital, and the labour share $\varepsilon_{h,t}$ equals the output elasticity for labour. Moreover, both factor shares sum to 1.

We consider two alternative specifications of the production function. The representative firm produces output according to either a VES or a CD production function. If the production function is VES, then output per worker is given by

$$y_t = [\alpha(q_t k_t)^{\omega\psi} (z_t h_t)^{(1-\omega)\psi} + (1 - \alpha)(z_t h_t)^{\psi}]^{1/\psi},$$

where $\psi \leq 1$, $0 < \alpha < 1$ and $0 < \omega < 1$. The VES production function might result from a CES production function $y_t = [\alpha(q_t u_t k_t)^{\psi} + (1 - \alpha)(z_t h_t)^{\psi}]^{1/\psi}$, where the utilization rate of capital is an increasing function of the ratio of efficient labour to efficient capital, $u_t = (q_t k_t/(z_t h_t))^{\omega-1}$. If the production function is CD, then detrended output per worker is given by

$$y_t = A_t k_t^{\varepsilon} h_t^{1-\varepsilon},$$

where $0 < \varepsilon < 1$, and A_t is the *efficiency wedge*. If the production function is VES, then output elasticity for capital is variable and given by

$$\varepsilon_t = \alpha\omega \left(\alpha + (1 - \alpha) \left(\frac{z_t h_t}{q_t k_t} \right)^{\omega\psi} \right)^{-1},$$

and if the production function is CD, then output elasticity for capital is constant and given by

$$\varepsilon_t = \varepsilon.$$

We remark that in general, the output elasticities for factors are functions of the capital–labour ratio (both factors adjusted by their efficiency). Therefore, assuming that the factor income shares equal the output elasticities for factors, it is generally possible to calibrate q_t/z_t . However, in the CD case, q_t and z_t cannot be identified because output elasticities for factors do not depend on the ratio q_t/z_t .

The resource constraint in terms of the detrended variables per capita is

$$c_t + x_t + g_t = y_t, \quad (11)$$

where $c_t = C_t/((1 + \gamma)^t L_t)$ is detrended consumption per capita, $x_t = X_t/((1 + \gamma)^t L_t)$ is detrended investment per capita, and $g_t = G_t/((1 + \gamma)^t L_t)$ is the *resource constraint wedge*.

The move law of detrended capital per capita is

$$(1 + \eta)(1 + \gamma)k_{t+1} = x_t + (1 - \delta)k_t - \frac{\phi}{2} \left(\frac{x_t}{k_t} - \kappa \right)^2 k_t, \quad (12)$$

where $0 < \delta < 1$ is the economic depreciation rate of capital, and capital accumulation includes quadratic adjustment costs $\phi > 0$ and $\kappa > 0$.

The representative household at time t is composed of L_t members. Each member of the representative household is endowed with one unit of time, which can be shared between leisure, $1 - h_t$, and labour, $0 < h_t < 1$, in return for a wage W_t . The intertemporal utility function of the representative household is

$$U_t = E_t \left\{ \sum_{t=0}^{\infty} L_t \beta^t [\log C_{L,t} + \mu \log(1 - h_t)] \right\},$$

where $C_{L,t} = C_t/L_t$ is consumption per capita, $0 < \beta < 1$ is the discount factor, and $\mu > 0$ is the value of leisure relative to consumption. The household budget constraint is

$$L_t C_{L,t} + \pi_{x,t} X_t = \pi_{h,t} W_t h_t L_t + r_t K_t + B_t,$$

where B_t are lump-sum transfers, $\pi_{h,t}$ is the *labour wedge*, and $\pi_{x,t}^{-1}$ is the *investment wedge*.

The first-order conditions characterizing a maximum of the household problem are

$$\frac{1 + \gamma}{\beta} \frac{1}{c_t} = E_t \left\{ \frac{1 + i_{t+1}}{c_{t+1}} \right\}, \quad (13)$$

$$r_{t+1} = \frac{\pi_{x,t}(1 + i_{t+1})}{1 - \phi((x_t/k_t) - \kappa)} + \frac{\pi_{x,t+1}}{1 - \phi((x_{t+1}/k_{t+1}) - \kappa)} \times \left[\frac{\phi}{2} \left(\frac{x_{t+1}}{k_{t+1}} - \kappa \right)^2 - \phi \left(\frac{x_{t+1}}{k_{t+1}} - \kappa \right) \frac{x_{t+1}}{k_{t+1}} - (1 - \delta_{t+1}) \right] \quad (14)$$

and

$$\mu \frac{c_t}{1 - h_t} = \pi_{h,t} W_t. \quad (15)$$

Here, i_{t+1} is the interest rate at time $t + 1$. Equation (13) is the Euler equation, according to which expected (discounted) marginal utilities are equal over time. Equation (14) establishes that the rental price of capital equals its user cost, which, in addition to the interest rate and the economic depreciation rate, also includes the investment wedge and the investment adjustment costs. Equation (15) states that the marginal rate of substitution between consumption and leisure equals the wage adjusted by the labour wedge.

The exogenous states (the wedges) follow a five- or four-dimensional vector that is autoregressive of order 1 where the error process is assumed to be multivariate normal with mean zero, and variance and covariance matrix $V = QQ'$, as described below:

$$s_{t+1} = P_0 + P s_t + Q v_{t+1} \quad (16)$$

in which $s_t = (\log q_t, \log z_t, \log \pi_{h,t}, \log \pi_{x,t}, \log g_t)$ in the VES case, and $s_t = (\log A_t, \log \pi_{h,t}, \log \pi_{x,t}, \log g_t)$ in the CD case.

The system of equations (7)–(16) characterizes the equilibrium of the economy.

TABLE 1 Model Parameters and BGP Variables (Annualized)

Parameter	Description	USA	Euro area 12
<i>Model parameters held constant across countries</i>			
δ	Depreciation rate of capital		0.0500
ψ	Production function parameter		-2.0000
ω	Production function parameter		0.5000
β	Discount factor		0.9750
μ	Relative value of leisure		2.5000
<i>Model parameters specific to each country</i>			
η	Population growth rate	0.090	0.0002
γ	Growth rate of output per worker	0.0198	0.0017
α	Production function parameter	0.8318	0.7526
ϕ	Adjustment cost parameter	12.5574	18.9162
<i>BGP variables</i>			
h	Hours worked per capita	0.2403	0.2503
k/y	Capital-output ratio	3.5261	4.7830
x/y	Investment rate	0.2808	0.2529
c/y	Consumption-to-output ratio	0.5931	0.5361
ε	Output elasticity for labour	0.5841	0.6253

4 | THE BUSINESS CYCLE ACCOUNTING PROCEDURE

Given the values for the parameters in Table 1, the model is solved for the steady-state quantities and the equilibrium is found. Equilibrium decision rules are derived assuming that the exogenous states follow the previous five- or four-dimensional vector that is autoregressive of order 1 where the matrices of autoregressive coefficients P_0 and P can be estimated.

The data are used as observables, and the Kalman filter is used to back out the wedges. The procedure involves: (i) solving the model for steady-state quantities; (ii) computing decision rules by log-linearization around the steady state; and (iii) building a state-space representation of the model, with a matrix for the laws of motion for the state variables, which are subject to Gaussian innovations, and a matrix with the optimal choices for output, hours, investment, government consumption and the output elasticity for labour as a function of the states. The likelihood of the innovations being jointly normal is computed, and the optimization program concerns the choice of the parameters of the vector autoregression (VAR), that is, the matrices P and Q , such that the likelihood is maximized. The vector P_0 is set such that $P_0 = (I - P)^{-1} E[s_t]$. We explain the choice of $E[s_t]$ in the next subsection. For a detailed list of different maximum likelihood estimation (MLE) methods, see Adjemian *et al.* (2022) and the references contained therein. Tables 2 and 3 present the estimated parameters of the VAR processes.

We extend the procedure proposed by Chari *et al.* (2007) in the wedge measurement step when the VES technology is assumed. Like them, we measure the government consumption wedge directly from the data, as explained in the ‘Data’ subsection below. To obtain the values of the other wedges, we use the data and the model’s decision rules. With y_t^d , h_t^d , x_t^d , g_t^d and ε_t^d denoting the data for the model variables and the observed labour share, $y(k_t, s_t)$, $h(k_t, s_t)$, $x(k_t, s_t)$ and $\varepsilon(k_t, s_t)$ denoting the decision rules of the model, and k_0^d being an initial condition for capital, the realized wedge series s_t^d solves the system given by

$$y_t^d = y(k_t^d, s_t^d), \quad h_t^d = h(k_t^d, s_t^d), \quad x_t^d = x(k_t^d, s_t^d), \quad \varepsilon_t^d = \varepsilon(k_t^d, s_t^d),$$

TABLE 2 Parameters of VAR(1) for the USA

Coefficient matrix P	Coefficient matrix Q ($V = QQ'$)
<i>Panel A: VES case</i>	
$\begin{pmatrix} 0.9300 & 0.1102 & 0.0425 & 0.0692 & 0.0699 \\ (0.0355) & (0.0953) & (0.1345) & (0.1718) & (0.0858) \\ 0.0472 & 0.9117 & -0.0594 & -0.1033 & -0.0256 \\ (0.0302) & (0.0830) & (0.1138) & (0.1446) & (0.0729) \\ 0.0335 & 0.0951 & 0.9300 & 0.0951 & -0.0254 \\ (0.0151) & (0.0287) & (0.0001) & (0.0480) & (0.0188) \\ 0.0070 & 0.0018 & -0.0493 & 0.9300 & 0.0049 \\ (0.0067) & (0.0114) & (0.0144) & (0.0163) & (0.0092) \\ -0.0132 & -0.1022 & -0.0640 & 0.0548 & 0.9300 \\ (0.0135) & (0.0291) & (0.0277) & (0.0479) & (0.0001) \end{pmatrix}$	$\begin{pmatrix} 0.0713 & 0 & 0 & 0 & 0 \\ (0.0060) & & & & \\ -0.0575 & 0.0093 & 0 & 0 & 0 \\ (0.0016) & (0.0052) & & & \\ -0.0050 & -0.0060 & 0.0191 & 0 & 0 \\ (0.0723) & (0.0745) & (0.0015) & & \\ 0.0012 & -0.0049 & 0.0025 & 0.0084 & 0 \\ (0.2311) & (0.2388) & (0.1756) & (0.0026) & \\ -0.0003 & -0.0033 & -0.0070 & 0.0021 & 0.0243 \\ (0.0899) & (0.0954) & (0.0585) & (0.1630) & (0.0020) \end{pmatrix}$
$E[s_t] = [-2.646, 1.434, -0.220, 0.204, -2.063]$	
<i>Panel B: CD case</i>	
$\begin{pmatrix} 0.8770 & 0.0108 & -0.0953 & 0.0297 \\ (0.0004) & (0.0123) & (0.0199) & (0.0070) \\ 0.0164 & 0.9820 & 0.0164 & -0.0138 \\ (0.0245) & (0.0003) & (0.0219) & (0.0052) \\ -0.0454 & -0.0117 & 0.9570 & -0.0050 \\ (0.0252) & (0.0117) & (0.0239) & (0.0070) \\ -0.1596 & -0.0839 & -0.1440 & 0.9480 \\ (0.0881) & (0.0312) & (0.0827) & (0.0006) \end{pmatrix}$	$\begin{pmatrix} 0.0095 & 0 & 0 & 0 \\ (0.0007) & & & \\ -0.0030 & 0.0086 & 0 & 0 \\ (0.0633) & (0.0007) & & \\ -0.0043 & -0.0038 & 0.0110 & 0 \\ (0.1075) & (0.1147) & (0.0022) & \\ -0.0017 & -0.0106 & 0.0044 & 0.0207 \\ (0.0734) & (0.0714) & (0.1127) & (0.0015) \end{pmatrix}$
$E[s_t] = [-0.450, -0.220, 0.204, -2.063]$	

Notes: Quarterly data 1979Q1 to 2020Q4. Numbers in parentheses are standard deviations of parameters estimated by MLE.

with $g_t = g_t^d$ and

$$(1 + \eta)(1 + \gamma)k_{t+1}^d = x_t^d + (1 - \delta)k_t^d - \frac{\phi}{2} \left(\frac{x_t^d}{k_t^d} - \kappa \right)^2 k_t^d,$$

given k_0^d .

We use these values for the wedges in our experiments. Finally, we perform simulations to see to what extent models with just one wedge or a combination of wedges can replicate observed data. Hence new decision rules are computed, setting the wedges, which are excluded in a specific simulation exercise, to their unconditional mean values throughout the simulation procedure. Since they are no longer random variables in the simulations, the computation of new equilibrium decision rules and allocations in the simulated economies ensures that the agent's expectations are consistent with the model.

4.1 | Parametrization and calibration

The parameters held fixed across the USA and the euro area are as follows: the annualized discount factor $\beta = 0.975$, the annualized depreciation rate $\delta = 0.05$, and the relative value of leisure

TABLE 3 Parameters of VAR(1) for the Euro Area 12

Coefficient matrix P					Coefficient matrix Q ($V = QQ'$)				
<i>Panel A: VES case</i>									
$\begin{pmatrix} -0.0708 & -0.6845 & 0.4101 & 1.0007 & 0.1755 \\ (0.3145) & (0.3164) & (0.3010) & (0.6047) & (0.1888) \\ -0.2891 & 0.6460 & -0.1479 & 0.0664 & -0.0056 \\ (0.2484) & (0.2901) & (0.2839) & (0.3260) & (0.0766) \\ -0.2539 & -0.0297 & 0.7869 & -0.0297 & -0.0063 \\ (0.1449) & (0.1565) & (0.1591) & (0.2272) & (0.0639) \\ -0.2385 & -0.1762 & -0.0951 & 0.9800 & -0.0028 \\ (0.0839) & (0.0885) & (0.1027) & (0.1174) & (0.0332) \\ 0.8512 & 0.5497 & -0.5305 & -1.2546 & 0.7933 \\ (0.4007) & (0.3864) & (0.3789) & (0.8775) & (0.2732) \end{pmatrix}$					$\begin{pmatrix} 0.0217 & 0 & 0 & 0 & 0 \\ (0.0022) & & & & \\ -0.0121 & 0.0173 & 0 & 0 & 0 \\ (0.0801) & (0.0023) & & & \\ -0.0021 & 0.0077 & 0.0087 & 0 & 0 \\ (0.1474) & (0.0921) & (0.0015) & & \\ 0.0039 & 0.0073 & 0.0007 & 0.0058 & 0 \\ (0.1561) & (0.1870) & (0.1699) & (0.0015) & \\ -0.0002 & 0.0080 & -0.0096 & 0.0157 & 0.0112 \\ (0.1549) & (0.1418) & (0.1635) & (0.1778) & (0.0028) \end{pmatrix}$				
$E[s_t] = [-2.957, 1.452, -0.408, 0.002, -1.593]$									
<i>Panel B: CD case</i>									
$\begin{pmatrix} 0.6445 & -0.2391 & 0.0116 & -0.0295 \\ (0.2695) & (0.3566) & (0.3481) & (0.0852) \\ 0.1029 & 0.7182 & 0.1029 & -0.0511 \\ (0.1238) & (0.1751) & (0.1962) & (0.0434) \\ -0.1000 & -0.1945 & 0.8314 & -0.0421 \\ (0.1056) & (0.1403) & (0.1604) & (0.0480) \\ -0.0492 & -0.2968 & -0.4019 & 0.9033 \\ (0.2601) & (0.2989) & (0.6105) & (0.1432) \end{pmatrix}$					$\begin{pmatrix} 0.0202 & 0 & 0 & 0 \\ (0.0017) & & & \\ 0.0069 & 0.0088 & 0 & 0 \\ (0.1041) & (0.0016) & & \\ 0.0065 & 0.0007 & 0.0068 & 0 \\ (0.1490) & (0.2363) & (0.0018) & \\ -0.0001 & -0.0131 & 0.0189 & 0.0121 \\ (0.1834) & (0.1527) & (0.3136) & (0.0030) \end{pmatrix}$				
$E[s_t] = [-0.332, -0.408, 0.002, -1.593]$									

Notes: Quarterly data 2005Q1 to 2020Q3. Numbers in parentheses are standard deviations of parameters estimated by MLE.

$\mu = 2.5$. We set $\psi = -2$ and $\omega = 0.5$, which are values near the del Río and Lores (2019) estimates. In particular, these authors estimate $\psi = -1.9$ and $\omega = 0.36$. We consider a value of ω higher than the estimated value above of del Río and Lores (2019) because to have well-defined series for z_t and q_t , it is necessary that $\omega > \varepsilon_t$ for all t . Other parameters are specific to the USA or the euro area: η is the average growth rate of population, γ is the average growth rate of output per capita, and ϕ and κ are the adjustment cost coefficients. We normalize so that detrended log output has a mean value of zero over the sample period.

If κ equals the investment–capital ratio along a balanced growth path (BGP), then it follows from equation (12) that $\kappa = (1 + \eta)(1 + \gamma) - (1 - \delta)$. The adjustment costs function is quadratic, which is typical in the macroeconomic literature and is also used in Brinca *et al.* (2016). To perform the quantitative analyses, we follow Brinca *et al.* (2016) and set $\phi = 0.25/\kappa$ to obtain the elasticity of the price of capital regarding the investment–capital ratio 0.25. Parameter α is set so that $\alpha\omega$ equals the sample average for capital share. Therefore when $\psi \rightarrow 0$, the VES production function converges to a CD production function with output elasticity for capital $\alpha\omega$.

The mean values of exogenous states $E[s_t]$ are calculated by solving the equation system (7)–(16) when $v_t = E[v_t] = 0$ to reproduce the sample averages for detrended hours worked, detrended output per capita, investment rate, consumption to output rate, and capital share. In the CD case, we calibrate a mean value for only one efficiency wedge, A . To compare the VES and CD economies, we choose the CD production function parameter ε equal to the output elasticity for capital in the BGP of the VES economy. Table 1 summarizes the parameters held constant across countries, and those that are specific to the USA and the euro area.

4.2 | Data

Here, we describe the data used in our quantitative analysis. The euro area is made up of 12 countries (Austria, Belgium, Finland, France, Germany, Greece, Ireland, Italy, Luxembourg, the Netherlands, Portugal and Spain). The value of a variable in the euro area is the sum of the values of that variable in the 12 countries. Data for the USA are provided by the National Income and Products Accounts (NIPA) and for Europe by Eurostat.

4.2.1 | Taxes: USA and euro area

We follow Prescott (2004) to compute taxes. We compute net direct taxes on personal expenditures in consumption and services, T_N^p , as a fraction of taxes on production and imports plus subsidies $T_N^p = \mu T_N$, where $\mu = \frac{2}{3} + \frac{1}{3}(G_N^d + G_N^{nds})/(G_N^d + G_N^{nds} + G_N^x)$, G_N^{nds} are nominal expenditures in non-durable consumption and services, G_N^d are nominal expenditures in durable consumer goods, and G_N^x is nominal Eurostat gross capital formation (resp. nominal NIPA gross private domestic investment). Durable consumer goods are excluded from consumption and included in investment, and taxes on non-durable consumption and services are then $T_N^c = T_N^p G_N^{nds}/(G_N^{nds} + G_N^d)$, while taxes on durable consumption are $T_N^d = T_N^p G_N^d/(G_N^{nds} + G_N^d)$. Net direct taxes on investment equal the taxes on production and imports plus subsidies less taxes on non-durable consumer goods and services: $T_N^x = T_N - T_N^c$.

4.2.2 | Services of durable consumer goods and nominal output: USA and euro area

The nominal stock of durable goods is computed using the perpetual method of inventory $K_{t+1}^d = G_{N,t}^d - T_{N,t}^d + (1 - \delta_d)K_t^d$, where $\delta_d = 0.0694$ and $K_0^d = (G_{N,0}^d - T_{N,0}^d)/(g_d + \delta_d)$, with g_d the average quarterly growth rate of $G_N^d - T_N^d$ during the sample period for the USA (resp. the euro area). The flow of nominal services of durable consumer goods is $S_N^d = (i + \delta_d)K^d$, where $i = 0.00985$ is the interest rate. Nominal output Y_N is nominal NIPA (resp. Eurostat) GDP, denoted GDP_N , less NIPA (resp. Eurostat) taxes on production and imports plus subsidies, T_N , plus the flow of services of durable consumer goods S_N^d : $Y_N = GDP_N - T_N + S_N^d$.

4.2.3 | Consumption and investment: USA and euro area

From the expenditures on non-durables and services, we subtract taxes, and following Brinca *et al.* (2016), we add the flow of services of durable consumer goods, whereupon nominal consumption is $C_N = G_N^{nds} - T_N^c + S_N^d$. Following Brinca *et al.* (2016), we add personal expenditures in durable consumer goods to investment, and take away taxes. Nominal investment in the euro area is $X_N = G_N^x - T_N^x + G_N^d$, and in the USA is $X_N = G_N^x + G_N^{gx} - T_N^x + G_N^d$, where G_N^{gx} is nominal NIPA government gross investment.⁵

4.2.4 | Population and labour: USA

We follow Cociuba *et al.* (2018) in measuring population and worked hours. Population L is the civilian non-institutional population aged 16–65 years (data from the Bureau of Labor Statistics) plus military personnel on active duty (data from the Defense Manpower Data Center, Office of

the Secretary of Defense, and US Department of Defense). Worked hours H are total persons at work aged 16 years and over multiplied by the average hours worked per week by total persons at work and by the number of weeks of a quarter, namely $52/4$. Series of persons at work and average hours worked per week are provided by the Bureau of Labor Statistics. Therefore worked hours per capita are H/L , and the fraction of time devoted to work, h , equals the worked hours per capita divided by the available hours per person (two-thirds of the total hours, 2184 hours per quarter, $H_A = 1456$), that is, $h = H/(LH_A)$.

4.2.5 | Population and labour: euro area

Population L is the Eurostat population aged 16–65 years for the 12 countries of the euro area. Worked hours are $H = 52(v_s L_s + v_e L_e)/4$, where L_s the Eurostat self-employed workers, L_e are Eurostat employees, v_s are Eurostat average week hours worked by a self-employed worker, v_e are Eurostat average week hours worked by employees, and $52/4$ is the number of weeks of a quarter. Here, v_s and v_e are the weighted averages of the average week hours in the 12 countries of the euro area, being the weights of the self-employed workers in the country and the employees in the country relative to the total self-employed workers and the total employees in the euro area. Therefore worked hours per capita are H/L , and the fraction of time devoted to work, h , equals the worked hours per capita divided by the available hours per person (two-thirds of the total hours, 2184 hours per quarter, $H_A = 1456$), that is, $h = H/(LH_A)$.

4.2.6 | Factor shares: USA

We construct the factor shares using an economy-wide definition standard in the macroeconomics literature (see Koh *et al.* 2020). Data are taken from NIPA. We split the proprietors' income into capital and labour incomes using the factor shares of the unambiguous income of the economy. In particular, with unambiguous capital income

$$\begin{aligned} UCI = S_N^d &+ \text{rental income} + \text{corporate profits} + \text{net interest} \\ &+ \text{current surplus government enterprises} \\ &+ \text{business current transfers payments} + \text{statistical discrepancy} \end{aligned}$$

and unambiguous income

$$UI = UCI + DEP + \text{compensation of employees},$$

where DEP is depreciation, we define the factor share of the unambiguous capital income as $\theta = (UCI + DEP)/(UI)$, and the ambiguous capital income as $ACI = \theta PI$, where PI is proprietors' income. Then the capital share is $\varepsilon_K = (UCI + DEP + ACI)/Y_N$, and $\varepsilon_L = 1 - \varepsilon_K$ is the labour share.

4.2.7 | Factor shares: euro area

The labour share is computed imputing to self-employed workers the same wage per hour as employees, $\varepsilon_L = W_N/Y_N(1 + v_s L_s/v_e L_e)$, where W_N is the Eurostat compensation of employees.

4.2.8 | Real variables: USA and euro area

Real output, real consumption and real investment are computed deflating the corresponding nominal magnitudes by the implicit deflator of GDP (provided by NIPA or Eurostat): $Y = Y_N/P$, $C = C_N/P$ and $X = X_N/P$.

5 | THE ECONOMIC RECESSIONS AND THE ROLE OF THE WEDGES

In this section, we focus on the role of the wedges in the two main recessions that the US economy faced after the Second World War, excepting the recent COVID-19 recession. The first is the recession in the late 1970s, which we call the 1982 recession, and the second is the Great Recession, which began in 2008. Furthermore, we also analyse the Great Recession in the euro area. The wedge-alone components together with the corresponding observed variables (output, labour and investment) are displayed in the panels of Figures 4–9. The values of the ϕ -statistics are displayed in Table 4. The ϕ -statistic is defined by Brinca *et al.* (2016) and is intended to capture how closely a particular component tracks the underlying variable.⁶ The ϕ -statistic for $y_{i,t}$ of wedge i is

$$\phi_i^y = \frac{1 / \sum_t (y_{i,t} - y_t)^2}{\sum_j (1 / \sum_t (y_{j,t} - y_t)^2)},$$

where $y_{i,t} \in \{h_{i,t}, y_{i,t}, x_{i,t}\}$ is the wedge-alone component of a variable due to wedge i , and $y_t \in \{h_t, y_t, x_t\}$ is the corresponding observed variable. The growth rates from the peak to the trough of worked hours of the variables and their wedge-alone components for the three recessions are displayed in Table 5.

5.1 | The US Great Recession

5.1.1 | CD case

The labour wedge was the main force driving the evolution of output during the US Great Recession (see Figure 4(c)). As shown in Table 5, between 2008Q1 and 2009Q4, the decline in the labour wedge accounted for around 79% of the fall in detrended output per capita. As shown in Table 4, the value of the ϕ -statistic of the labour wedge for output is 0.60. The investment wedge also played a significant but secondary role in accounting for the evolution of output. Its ϕ -statistic for output is 0.24, and between 2008Q1 and 2009Q4, its decline accounted for 50% of the fall in detrended output per capita (see Table 5). During the US Great Recession, the main force driving the evolution of hours worked per capita was the labour wedge (its ϕ -statistic for labour is 0.94), while the main force driving the evolution of detrended investment per capita was the investment wedge (its ϕ -statistic was 0.68). However, the efficiency wedge played a negligible role; as shown in Table 4, the values of the ϕ -statistics of the efficiency wedges for labour, output and investment are 0.01, 0.11 and 0.07.

5.1.2 | VES case

The main force driving the output growth slowdown during the US Great Recession between 2008Q1 and 2009Q4 was the decline of the capital-efficiency wedge (see Figure 5(c)). As shown in

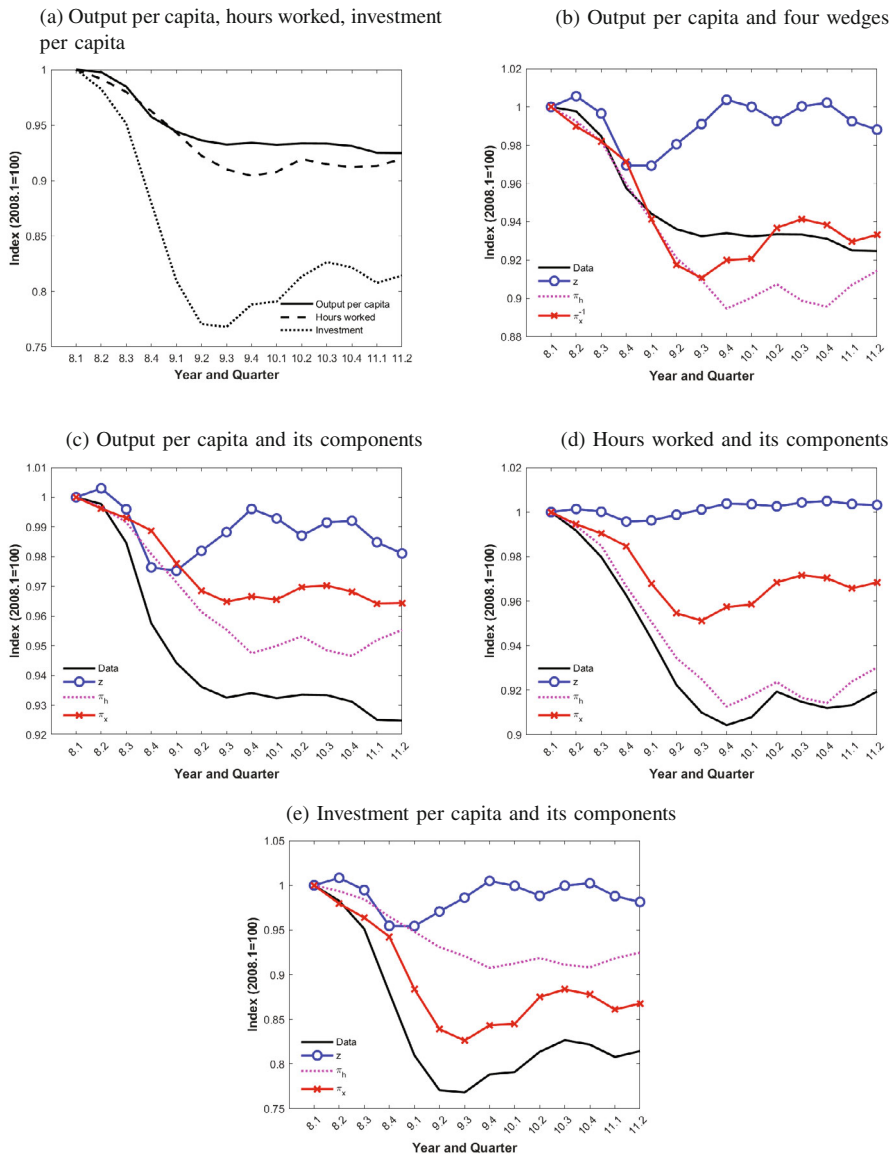
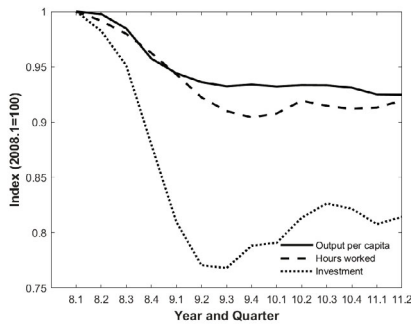


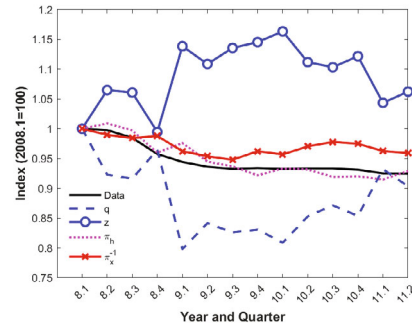
FIGURE 4 The US Great Recession with CD production function.

Table 5, the fall in the capital-efficiency wedge accounted for 155% of the decline in detrended output per capita between 2008Q1 and 2009Q4, while the drops of the investment and labour wedges accounted for 32% and 55%, respectively. The ϕ -statistic for output of the capital-efficiency wedge is 0.34, higher than the values of the ϕ -statistics for output of the other wedges (see Table 4). The contributions of the investment and labour wedges to the recession in terms of the evolution of output and investment were similar (see Figure 5(c)), but the drop of the labour wedge had a higher impact on the fall of hours worked (see Figures 5(d) and 5(e)). The fall in the investment (resp. labour) wedge between 2008Q1 and 2009Q4 accounted for 16% (resp. 65%) of the decline in hours worked per capita, 32% (resp. 55%) of the decline in detrended output per capita, and 31% (resp. 34%) of the decline in detrended investment per capita (see Table 5). As shown in Table 4, the ϕ -statistics for labour, output and investment of the labour (resp. investment) wedge are 0.61,

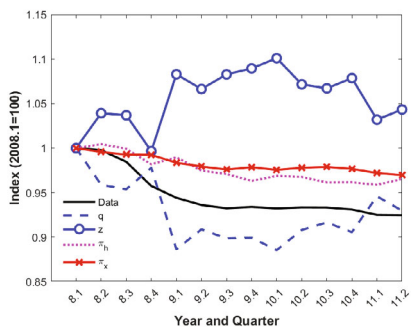
(a) Output per capita, hours worked, investment per capita



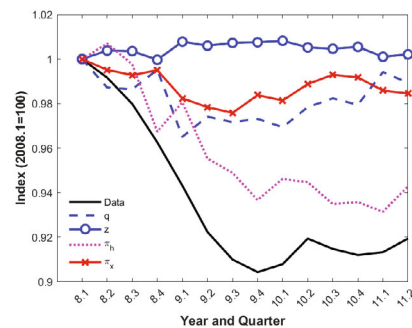
(b) Output per capita and four wedges



(c) Output per capita and its components



(d) Hours worked and its components



(e) Investment per capita and its components

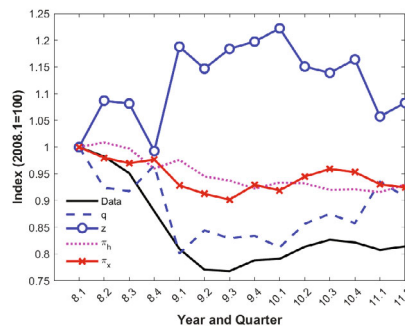


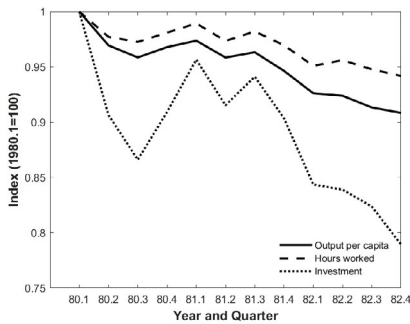
FIGURE 5 The US Great Recession with VES production function.

0.33 and 0.16 (resp. 0.12, 0.22 and 0.17). As shown in Figure 5 and Table 5, the countercyclical evolution of the labour-efficiency wedge contributed to reducing the fall of labour, output and investment during the US Great Recession.

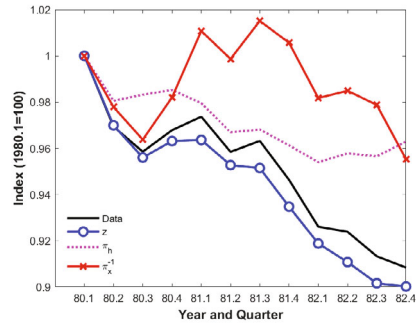
5.1.3 | Comparing the specifications

In the VES case compared to the CD case, the role of the labour wedge is reduced. Therefore it does not play a prominent role, but plays a secondary role in accounting for the evolution of output, and the degrees of importance of the labour and investment wedges become quite similar. Furthermore, in the CD case, the role played by the fall in the efficiency wedge is negligible, but

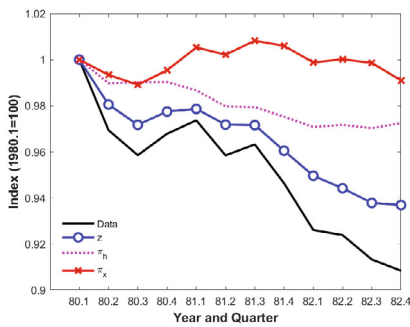
(a) Output per capita, hours worked, investment per capita



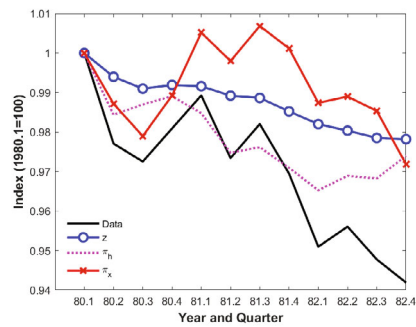
(b) Output per capita and four wedges



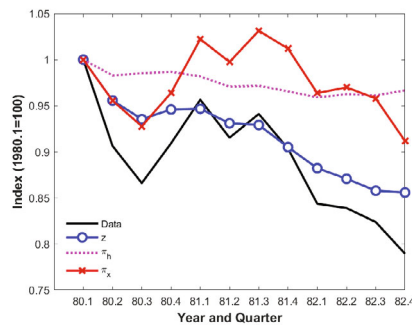
(c) Output per capita and its components



(d) Hours worked and its components



(e) Investment per capita and its components

**FIGURE 6** The US 1982 recession with CD production function.

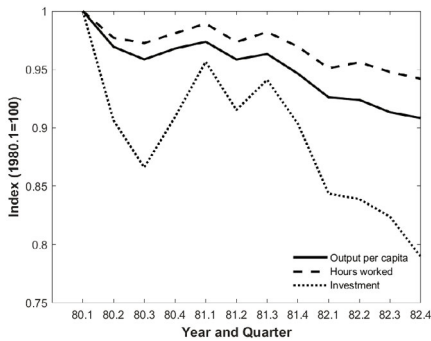
the VES case reveals that this result might be due to the labour and capital-efficiency wedges moving in opposite directions and cancelling each other out.

5.2 | The 1982 recession

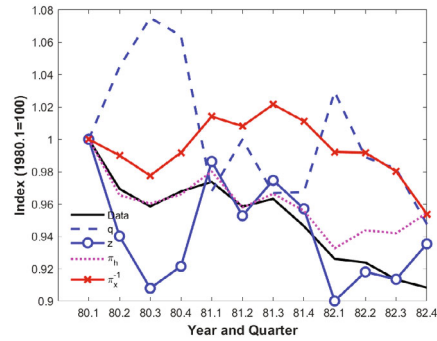
5.2.1 | CD case

The main force driving the evolution of output and investment during the US 1982 recession was the efficiency wedge (see Figures 6(c) and 6(e)). As shown in Table 4, the ϕ -statistic of the

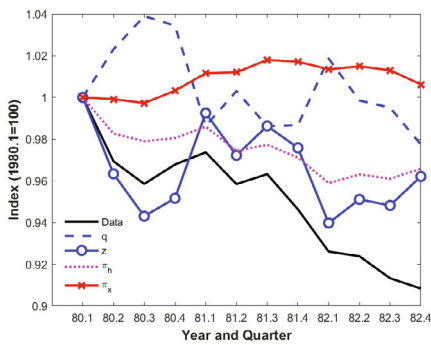
(a) Output per capita, hours worked, investment per capita



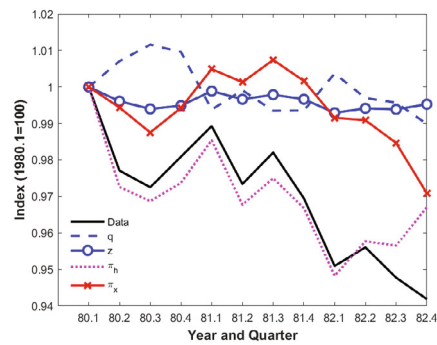
(b) Output per capita and four wedges



(c) Output per capita and its components



(d) Hours worked and its components



(e) Investment per capita and its components

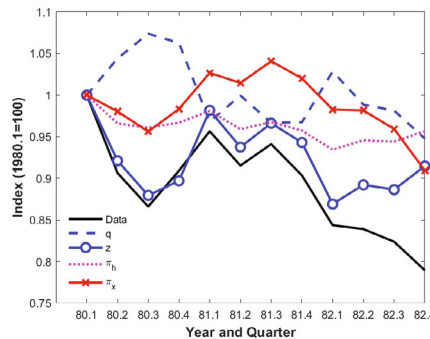
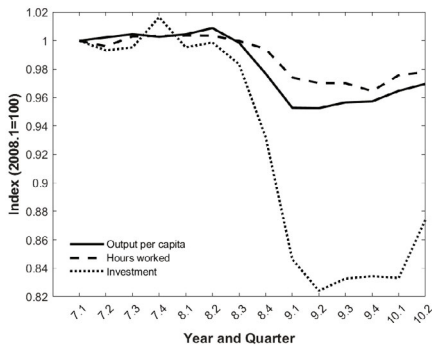


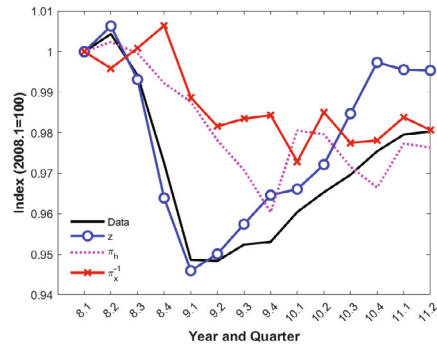
FIGURE 7 The US 1982 recession with VES production function.

efficiency wedge for output is 0.72, and for investment is 0.71, higher than the values of the ϕ -statistics for output and investment of the other wedges. The drop in the efficiency wedge accounted for 68% and 66% of the falls in detrended output per capita and detrended investment per capita (see Table 5). The labour wedge played a prominent role in accounting for the evolution of hours worked per capita. As shown in Table 4, its ϕ -statistic for labour is 0.54. The investment wedge played a negligible role in accounting for the evolution of labour, output and investment during the US 1982 recession (see Figures 6(c), 6(d) and 6(e)). Its ϕ -statistics for labour, output and investment are 0.16, 0.07 and 0.12 (see Table 4).

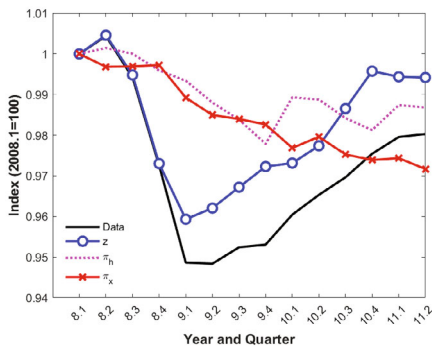
(a) Output per capita, hours worked, investment per capita



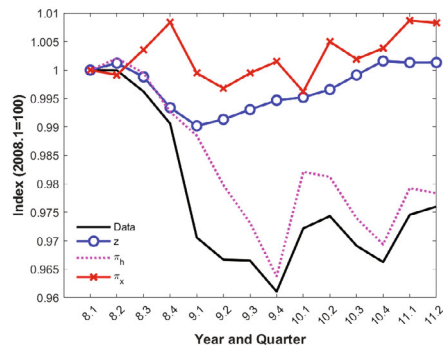
(b) Output per capita and four wedges



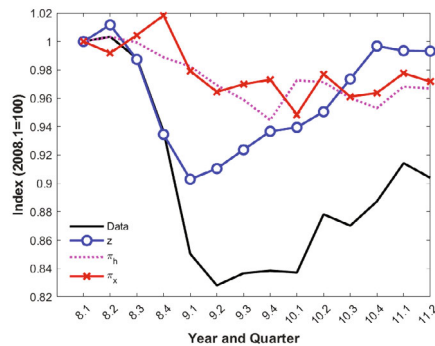
(c) Output per capita and its components



(d) Hours worked and its components



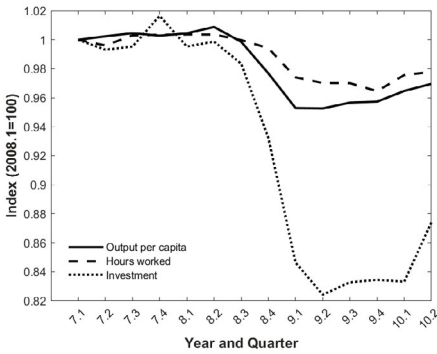
(e) Investment per capita and its components

**FIGURE 8** The euro area 12 Great Recession with CD production function.

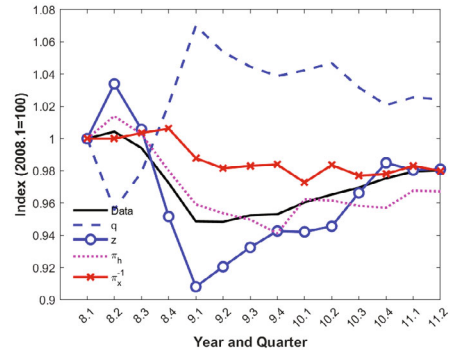
5.2.2 | VES case

The labour-efficiency wedge was the main force driving the evolution of output and investment during the US 1982 recession (see Figure 7(c)). In particular, the drop in the labour-efficiency wedge accounted for 40% and 38% of the falls in detrended output per capita and detrended investment per capita between 1980Q1 and 1982Q4 (see Table 5). The ϕ -statistic of the labour-efficiency wedge for output is 0.44, and for investment 0.55 (see Table 4). The labour

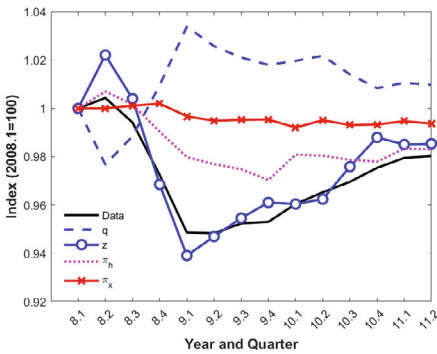
(a) Output per capita, hours worked, investment per capita



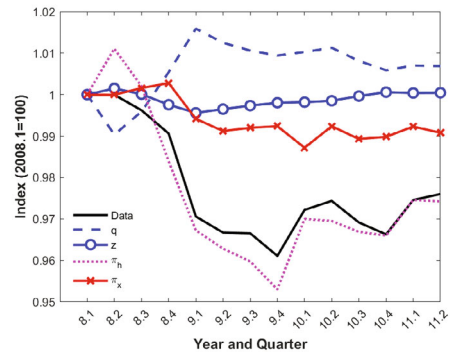
(b) Output per capita and four wedges



(c) Output per capita and its components



(d) Hours worked and its components



(e) Investment per capita and its components

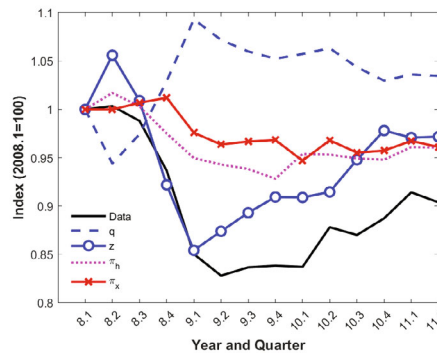


FIGURE 9 The euro area 12 Great Recession with VES production function.

wedge played a prominent role in accounting for the evolution of hours worked (its ϕ -statistic for labour is 0.76), and a significant but secondary role in accounting for the evolution of output (its ϕ -statistic for output is 0.33). The investment and capital-efficiency wedges played negligible roles in accounting for the evolution of the variables during the US 1982 recession. The ϕ -statistics of the investment (resp. capital-efficiency) wedges for labour, output and investment are 0.08, 0.07 and 0.11 (resp. 0.05, 0.08 and 0.07).

TABLE 4 ϕ -statistics

Variable	VES					CD			
	ϕ_q^y	ϕ_z^y	$\phi_{\pi_h}^y$	$\phi_{\pi_x}^y$	ϕ_g^y	ϕ_A^y	$\phi_{\pi_h}^y$	$\phi_{\pi_x}^y$	ϕ_g^y
<i>US Great Recession</i>									
<i>h</i>	0.14	0.07	0.61	0.12	0.06	0.01	0.94	0.04	0.01
<i>y</i>	0.34	0.02	0.33	0.22	0.08	0.11	0.60	0.24	0.06
<i>x</i>	0.56	0.02	0.16	0.17	0.09	0.07	0.17	0.68	0.08
<i>US 1982 recession</i>									
<i>h</i>	0.05	0.06	0.76	0.08	0.04	0.23	0.54	0.16	0.07
<i>y</i>	0.08	0.44	0.33	0.07	0.08	0.72	0.15	0.07	0.06
<i>x</i>	0.07	0.55	0.18	0.11	0.09	0.71	0.11	0.12	0.07
<i>Euro area 12 Great Recession</i>									
<i>h</i>	0.02	0.04	0.86	0.06	0.02	0.09	0.82	0.06	0.04
<i>y</i>	0.02	0.73	0.17	0.05	0.03	0.56	0.17	0.21	0.06
<i>x</i>	0.05	0.43	0.24	0.16	0.12	0.35	0.25	0.23	0.17

TABLE 5 Wedge-alone Components, Deviation from Peak (Hours Trough)

Data		VES					CD			
		$\mathbf{y}_{q,t}$	$\mathbf{y}_{z,t}$	$\mathbf{y}_{\pi_h,t}$	$\mathbf{y}_{\pi_x,t}$	$\mathbf{y}_{g,t}$	$\mathbf{y}_{A,t}$	$\mathbf{y}_{\pi_h,t}$	$\mathbf{y}_{\pi_x,t}$	$\mathbf{y}_{g,t}$
US Great Recession										
h	−10.06	−2.72	0.75	−6.54	−1.63	1.94	0.38	−9.14	−4.36	2.22
y	−6.82	−10.59	8.58	−3.73	−2.15	0.83	−0.39	−5.40	−3.40	1.01
x	−23.79	−18.13	18.02	−8.08	−7.34	−2.17	0.49	−9.71	−17.05	−1.81
US 1982 recession										
h	−5.99	−1.02	−0.47	−3.35	−2.96	0.24	−2.21	−2.64	−2.85	0.28
y	−9.61	−2.29	−3.86	−3.48	0.62	0.01	−6.52	−2.79	−0.91	0.06
x	−23.65	−5.37	−8.92	−4.43	−9.49	−0.23	−15.58	−3.39	−9.25	−0.21
Euro area 12 Great Recession										
h	−3.40	1.06	−0.27	−4.11	−0.80	0.90	−0.70	−2.72	−0.05	1.09
y	−4.88	2.09	−4.64	−2.55	−0.48	0.54	−3.33	−1.63	−1.62	0.68
x	−17.83	5.82	−11.31	−6.36	−3.37	−2.11	−7.94	−4.19	−3.06	−1.64

5.2.3 | Comparing the specifications

The labour wedge gains prominence in the VES case compared to the CD case. The VES case reveals that the fall in the efficiency wedge is driven mainly by the labour-efficiency wedge.

5.3 | The Great Recession in the euro area

5.3.1 | CD case

The main force driving the evolution of output and investment during the euro area Great Recession was the efficiency wedge (see Figures 8(c) and 8(e)). As shown in Table 4, the ϕ -statistic

of the efficiency wedge for output is 0.56, and for investment is 0.35, higher than the values of the ϕ -statistics for output and investment of the other wedges. The drop in the efficiency wedge accounted for 68.2% and 44.5% of the falls in detrended output per capita and detrended investment per capita between 2008Q1 and 2009Q4 (see Table 5). The investment and labour wedges played significant but secondary roles in accounting for the evolution of output. The ϕ -statistic of the investment (resp. labour) wedge for output is 0.21 (resp. 0.17); see Table 4. The labour wedge played a prominent role in accounting for the evolution of hours worked per capita. As shown in Table 4, its ϕ -statistic for labour is 0.82.

5.3.2 | VES case

The labour-efficiency wedge was the main force driving the evolution of output and investment during the euro area Great Recession (see Figures 9(c) and 9(e)). As shown in Table 4, its ϕ -statistic for output is 0.73, and 0.43 for investment. The labour wedge played a significant but secondary role in accounting for the evolution of output and investment (its ϕ -statistic is 0.17 for output and 0.24 for investment) and a prominent role in accounting for the evolution of labour (its ϕ -statistic for labour is 0.86). The investment wedge played a negligible role in accounting for the evolution of output and labour, but it played a more significant role in accounting for the evolution of investment (see Figures 9(c), 9(d) and 9(e)). The capital-efficiency wedge displayed a countercyclical behaviour. It played a negligible role in accounting for the evolution of labour, output and investment during the euro area Great Recession (see Figures 9(c), 9(d) and 9(e)). As shown in Table 4, the ϕ -statistics of the investment (resp. capital-efficiency) wedges for labour, output and investment are 0.06, 0.05 and 0.16 (resp. 0.02, 0.02 and 0.05).

5.3.3 | Comparing the specifications

In the VES case compared to the CD case, the role of the labour wedge is strengthened, and the role of the investment wedge is reduced. Furthermore, the VES case reveals that the fall in the efficiency wedge is driven mainly by the labour-efficiency wedge.

6 | CONCLUSION

We conduct BCA exercises for the euro area and the USA under two alternative hypotheses. On the one hand, we assume that movements of the factor shares are driven by market frictions or non-competitive forces reflected in the labour and investment wedges. This is the standard assumption in the BCA literature. On the other hand, we assume that movements in the factor shares are driven by the frictionless competitive adjustment of the productive factors. We modify the Chari *et al.* (2007) BCA method to compute two efficiency wedges, one for capital and another for labour. In particular, to compute both efficiency wedges, we require the model to be consistent with the trend and cyclical behaviour of the factor income shares. We have computed the contribution of both efficiency wedges to the evolution of labour, output and investment during the Great Recession (in both the euro area and the USA) and the US 1982 recession.

If movements in factor shares are driven by market frictions or non-competitive forces, our results broadly confirm previous findings by Brinca *et al.* (2016). In particular, we find that the labour wedge played a prominent role in accounting for the evolution of output in the US Great Recession, whereas the investment wedge played a significant but secondary role, and the role played by the efficiency wedge was negligible. However, we find that the efficiency wedge was the

main force driving the evolution of output in the euro area Great Recession and the US 1982 recession.

If movements in factor shares are driven by the frictionless competitive adjustment of factors, then the overall picture sharply changes. In particular, we find that during the US Great Recession, the capital-efficiency wedge played a prominent role in accounting for the output growth slowdown, whereas the countercyclical evolution of the labour-efficiency wedge contributed to mitigating it, and the labour and investment wedges played significant but secondary roles. However, we find that the prominent force driving the output growth slowdown in both the euro area Great Recession and the US 1982 recession was the labour-efficiency wedge, whereas the labour wedge was a significant but secondary force, and the investment wedge a negligible one.

Our results have three main implications. First, they show that the same evolution of the efficiency wedge might conceal very different behaviours of the capital-efficiency and labour-efficiency wedges, hence just analysing the impact of the efficiency wedge on the macroeconomic variables might lead to a poor understanding of the economic recessions. In particular, the negligible role played by the efficiency wedge in accounting for the evolution of output during the US Great Recession found by previous works might be explained by the movement in opposite directions of both the labour and capital-efficiency wedges. Second, we find that the capital-efficiency wedge played a prominent role in accounting for the output growth slowdown during the US Great Recession, whereas the role played by the investment wedge was significant but secondary, which means that the frictions in the capital markets of the kind proposed by the so-called *investment friction theory* that produce investment-driven downturns in output might have been a prominent force driving the US Great Recession. Finally, our results suggest that the US Great Recession may have involved some propagation mechanisms significantly different from those working during the euro area Great Recession and the US 1982 recession, because in the US Great Recession, in addition to the significant role played by the investment wedge, the efficiency wedges moved in opposite directions to how they did in the other two recessions.

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ENDNOTES

- ¹ del Río and Lores (2021) find that the productivity slowdowns of the US economy in the 1970s and after the end of the 1990s were driven by the drop in the capital-efficiency wedge.
- ² As discussed in Section II, the drop of the capital-efficiency wedge is consistent with the observed decline of the prices of houses and capital assets during the US Great Recession.
- ³ Christiano *et al.* (2015) assert that the vast bulk of movements in economic activity during the Great Recession were due to financial frictions.
- ⁴ We have built the US relative housing price deflating the median sales price of houses sold for the USA (provided by the Federal Reserve Bank of St Louis) by the implicit deflator of GDP. We use annual data from the National Income and Product Accounts (NIPA) to build the relative price of US capital assets. First, we have built a price index of the capital assets dividing the NIPA current-cost net stock of fixed assets and consumer durable goods by the NIPA chain-type quantity indexes for net stock of fixed assets and consumer durable goods. Second, we have divided this index by the implicit deflator of GDP.
- ⁵ Under our definitions of consumption and investment, the resource constraint wedge g equals government consumption plus net exports (both per capita and detrended).
- ⁶ The ϕ -statistic has the desirable features that it lies in $[0, 1]$ and sums to 1 across the four wedges; and when a particular output component tracks a variable perfectly, its value is 1.

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